

1 Questions

Nick's Question 1.1
In the Wave equation paper you showed that for the Greens function, we have that

$$\mathcal{F}G(t, \cdot)(\xi) = \frac{\sin(t|\xi|)}{|\xi|}$$

and so we see that

$$\mathcal{F}G(t, \cdot)(c\xi) = \frac{1}{c} \mathcal{F}(ct, \cdot)(\xi)$$

However in the Fractional SPDE paper the greens function is more complicated and we have that

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} H_{1,2}^{1,1} \left(2^{-1} \nu t^\beta |\xi|^\alpha \middle| \begin{matrix} (0, 1) \\ (0, 1) \end{matrix} \right. (1 - \beta - \gamma, \beta) \Big) \\ &= t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_1^{1,1}(s) \left(2^{-1} \nu t^\beta |\xi|^\alpha \right)^{-s} ds \end{aligned}$$

And so we see that

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \frac{1}{2\pi i} \int_L \mathcal{H}_1^{1,1}(s) \left(2^{-1} \nu t^\beta |c\xi|^\alpha \right)^{-s} ds$$

However I am unsure how to handle the constant, c , on the inside so we can get a nice scaling property like we do in the Wave Equation paper.

Also, instead of using the Fox-H function we can just say

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(\xi) &= t^{\beta+\gamma-1} E_{\beta, \beta+\gamma}(-2^{-1} \nu t^\beta |\xi|^\alpha) \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

and so

$$\mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) = t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))}$$

but I still don't see how to handle the c .

Here is how one should proceed:

$$\begin{aligned} \mathcal{F}Y_{\alpha, \beta, \gamma, d}(t, \cdot)(c\xi) &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta |c\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu t^\beta c^\alpha |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= t^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \\ &= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} (tc^{\frac{\alpha}{\beta}})^{\beta+\gamma-1} \sum_{k=0}^{\infty} \frac{(-2^{-1} \nu (tc^{\frac{\alpha}{\beta}})^\beta |\xi|^\alpha)^k}{\Gamma(\beta k + (\beta + \gamma))} \end{aligned}$$

$$= c^{\frac{\alpha}{\beta}(1-\gamma-\beta)} \mathcal{F}Y_{\alpha,\beta,\gamma,d} \left(c^{\frac{\alpha}{\beta}} t, \cdot \right) (c\xi)$$

Please check if this scaling reduces to heat and wave equations correctly.

Nick's Question 1.2 In the wave equation paper with Xia Chen you said that that fundamental solution to the SPDE

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) - \frac{1}{2} \Delta u(t, x) = \sqrt{\Theta} u(t, x) \dot{W}(x) & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 \\ u_t(0, \cdot) = 0 \end{cases} \quad (1.1)$$

wspde

and the mild solution is defined to be

$$u(t, x) = 1 + \sqrt{\Theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \quad (1.2)$$

weqs

but for general initial conditions for the wave equation we have the solution to be

$$\begin{aligned} u(t, x) = & \frac{\partial}{\partial t} \int_{\mathbb{R}^d} G(t-s, x-y) u(0, y) dy + \int_{\mathbb{R}^d} G(t-s, x-y) u_t(0, y) dy \\ & + \sqrt{\Theta} \int_0^t \left(\int_{\mathbb{R}^d} G(t-s, x-y) u(s, y) W(\delta y) \right) ds \end{aligned} \quad (1.3)$$

gweqs

But when the initial conditions are such as in 1.1 then the first two integral in 1.3 are 1 and 0 respectively and so we arrive at 1.2.

In the fractional SPDE

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\rho(u(t, x)) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = \mu & \beta \in (0, 1] \\ u(0, \cdot) = \mu_0, \quad u_t(0, \cdot) = \mu_1 & \beta \in (1, 2] \end{cases} \quad (1.4)$$

fspde

and we define

$$J_0(t, x) = \begin{cases} (Z(t, \cdot) * \mu)(x) & \beta \in (0, 1] \\ (Z^*(t, \cdot) * \mu_0)(x) + (Z(t, \cdot) * \mu_1)(x) & \beta \in (1, 2] \end{cases}$$

and

$$I(t, x) = \iint_{[0,t] \times \mathbb{R}} Y(t-s, x-y) \rho(u(s, y)) W(\delta y) ds$$

and the general solution for 1.4 is

$$u(t, x) = J_0(t, x) + I(t, x) \quad (1.5)$$

feqs

My question is that if we consider the following fractional spde

$$\begin{cases} (\partial^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2}) u(t, x) = I_t^\gamma \left[\rho(u(t, x)) \dot{W}(x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = 1 & \beta \in (0, 1] \\ u(0, \cdot) = 1, \quad u_t(0, \cdot) = 0 & \beta \in (1, 2] \end{cases} \quad (1.6)$$

fspde1

then can we calculate $J_0(t, x)$? More specifically is $J_0(t, x) = 1$ in this case? The wave equation paper worked out nicely because in that paper $J_0(t, x) = 1$ and so the solution was able to be written as 1.2. The 1 allows us to easily do the Piccard iteration to get equation (2.3) from the Wave equation paper with Xia Chen.

So if our solution can not be written as

$$u(t, x) = 1 + I(t, x)$$

I am not sure at the moment how we can come up with an expression for the f'_n s in the Weiner Expansion like you did in equation (2.3) in the paper with Xia Chen.

Hi, Nick, here is the answer:

Notice that, by Theorem 4.1 of my paper,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}(t, \cdot)(0) \\ &= t^{\lceil \beta \rceil - 1} E_{\beta, \lceil \beta \rceil}(0) \\ &= \frac{t^{\lceil \beta \rceil - 1}}{\Gamma(\lceil \beta \rceil)}. \end{aligned}$$

Hence, when $\beta \in (0, 1]$, we see that $(Z_{\alpha, \beta, \gamma, d}(t, \cdot) * 1)(x) = 1$. When $\beta \in (1, 2]$,

$$\begin{aligned} (Z_{\alpha, \beta, \gamma, d}^*(t, \cdot) * 1)(x) &= \mathcal{F}Z_{\alpha, \beta, \gamma, d}^*(t, \cdot)(0) \\ &= E_{\beta}(0) \\ &= \frac{1}{\Gamma(1)} = 1. \end{aligned}$$

Hence, under the initial conditions given by (1.6), $J_0(t, x) \equiv 1$.

Nick's Question 1.3 Guess for the constant (1.20) from the Wave Equation Paper In the wave equation paper, the constant (1.20) was

$$C = \begin{cases} \frac{4 - \Delta}{3 - \Delta} & \text{under Assumption A} \\ \frac{4 - d}{3 - d} & \text{under Assumption B or C} \end{cases} \quad (1.7) \quad \boxed{\text{const}}$$

I believe in the fractional paper the constant will be

$$\begin{cases} \frac{2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha}{2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha} & \text{under Assumption A} \\ \frac{2(\beta + \gamma - 1) + 2 - \beta d/\alpha}{2(\beta + \gamma - 1) + 1 - \beta d/\alpha} & \text{under Assumption B or C} \end{cases}$$

I think this because in the wave equation paper where we use $(4 - \Delta)$ this gets replaced by $(2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha)$ and thus we can guess that

$$\begin{cases} 4 - \Delta & \rightarrow 2(\beta + \gamma - 1) + 2 - \beta\Delta/\alpha \\ 3 - \Delta & \rightarrow 2(\beta + \gamma - 1) + 1 - \beta\Delta/\alpha \end{cases}$$

Note that when $\alpha = \beta = 2$ and when $\gamma = 0$ then the guess reduces to the wave equation.

Nick's Question 1.4 Ignore, This is not correct In order to come up with 1.7, we need a version of Lemma 4.3 from the wave equation paper. I was think that we could just use this Lemma directly in the following way. First the Lemma says that

Lemma 1.1. *Under Assumption A with $0 < \Delta < d \leq 3$, it holds that*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((4 - \Delta_0)n + 1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[2^{4 - \Delta_0} 2^{-\Delta_0/2} \mathcal{M}^{\frac{4 - \Delta_0}{2}} \right] \quad (1.8)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{4 - \Delta_0} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{4 - \Delta_0} \right) 2^{4 - \Delta_0} 2^{-\Delta_0/2} \mathcal{M}^{\frac{4 - \Delta_0}{2}} \right] \quad (1.9)$$

Both relations hold also under wither Assumption B or Assumption C with α and $\mathcal{M}$ replaced by d and $\mathcal{M}(\delta_0)$ respectively.

What I was thinking was that the Δ_0 in the above lemma only needs to satisfy $0 < \Delta_0 < d \leq 3$. So if we set Δ_0 to be

$$\Delta_0 = \frac{\beta \Delta}{\alpha} + 2 - 2(\beta + \gamma - 1) = \frac{\beta \Delta}{\alpha} + 2(2 - \beta - \gamma)$$

Then

$$4 - \Delta_0 = 2(\beta + \gamma - 1) + 2 - \frac{\beta \Delta}{\alpha} = 2(\beta + \gamma) - \frac{\beta \Delta}{\alpha}$$

and by the above lemma this world mean we get the two equalities

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left(\Gamma((2(\beta + \gamma) - \frac{\beta \Delta}{\alpha})n + 1) \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[2^{2(\beta + \gamma) - \frac{\beta \Delta}{\alpha}} 2^{-(\frac{\beta \Delta}{\alpha} + 2(2 - \beta - \gamma))/2} \mathcal{M}^{\frac{2(\beta + \gamma) - \frac{\beta \Delta}{\alpha}}{2}} \right] \quad (1.10)$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log \left((n!)^{2(\beta + \gamma) - \frac{\beta \Delta}{\alpha}} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2 \right) = \log \left[\left(\frac{2}{2(\beta + \gamma) - \frac{\beta \Delta}{\alpha}} \right) 2^{2(\beta + \gamma) - \frac{\beta \Delta}{\alpha}} 2^{-(\frac{\beta \Delta}{\alpha} + 2(2 - \beta - \gamma))/2} \mathcal{M}^{\frac{2(\beta + \gamma) - \frac{\beta \Delta}{\alpha}}{2}} \right] \quad (1.11)$$

So in order to say this then we would need

$$0 < \Delta_0 < d \leq 3$$

which means we would need

$$0 < \frac{\beta \Delta}{\alpha} + 2(2 - \beta + \gamma) < d \leq 3$$

which implies that we need

$$\beta + \gamma - \frac{d}{2} < 2 + \frac{1}{2} \left(\frac{\beta \Delta}{\alpha} - d \right) < \beta + \gamma$$

I am not sure how restrictive this is yet and maybe Dalang's condition says this is ok.

Nick's Question 1.5 Lemma 2.3 for the fractional equation

1. In the outline you sent me, you use

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

Where $H^a(\mathbb{R}^d)$ is the fractional Sobolev Space where $H^2(\mathbb{R}^d) = W^{1,2}(\mathbb{R}^d)$. However, I saw on line that $H^a(\mathbb{R}^d) = W^{a,2}(\mathbb{R}^d)$ and so $H^2(\mathbb{R}^d) = W^{2,2}(\mathbb{R}^d)$. Is there a mistake here?

2. Where is $\mathcal{M}$ defined with the sup over $\mathcal{F}_d$ and not $\mathcal{F}_a$? Is this a typo?
3. When defining $\mathcal{E}_a(g, g)$ why does the $(2\pi)^{-d}$ appear here but not in the wave paper?
4. How do we generalize ρ from the wave paper to the fractional equation? For the wave equation we have that

$$\rho = (\mathcal{M}(\gamma, 2))^{(4-\alpha)/2}$$

so it makes sense that it should generalize to

$$\rho = (\mathcal{M}_a(\gamma, 2))^{(2a-\alpha)/a} = 2^{-a/\alpha} (\mathcal{M}_a(\gamma, 1))^{(2a-\alpha)/a}$$

but showing the first equality seems hard.

2 Proof Sketches

Nick's Question 2.1 Scaling Property Suppose that $\mu(d\xi) = \varphi(\xi)d\xi$ and φ satisfies $\varphi(c\xi) = c^{-(d-\Delta)}$ for some $\Delta \in (0, d)$ and for any $c > 0$. Also recall that

$$\mathcal{F}Y(ct, \cdot)(\xi) = c^{\beta+\gamma-1} \mathcal{F}Y(t, \cdot)(t^{\beta/\alpha}\xi).$$

Then

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = t^{[2(\beta+\gamma-1)+2-\beta\Delta/\alpha]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2$$

Proof:

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{H^{\otimes n}}^2 = \frac{1}{n!} \sum_{\rho \in \Sigma_n} \int_{\mathbb{R}^{nd}} \mathcal{F}f_n(\cdot, 0, t)^2(\vec{\xi}) \varphi(\vec{\xi}) d\vec{\xi}$$

where $\varphi(\vec{\xi}) = \varphi(\xi_1) \cdots \varphi(\xi_n)$

$$\begin{aligned} \int_{\mathbb{R}^{nd}} \mathcal{F}f_n(\cdot, 0, t)^2 \varphi(\vec{\xi}) d\vec{\xi} &= \int_{\mathbb{R}^{nd}} \left(\int_{[0,t]^{n <}} \overline{\mathcal{F}Y(t-t_n, \cdot)(\xi_1 + \cdots + \xi_n) \cdots \mathcal{F}Y(t_2 - t_1, \cdot)(\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\ &= t^{2n(\beta+\gamma-1)} \int_{\mathbb{R}^{nd}} \left(\int_{[0,t]^{n <}} \overline{\mathcal{F}Y\left(1 - \frac{t_n}{t}, \cdot\right) (t^{\beta/\alpha}(\xi_1 + \cdots + \xi_n)) \cdots \mathcal{F}Y\left(\frac{t_2}{t} - \frac{t_1}{t}, \cdot\right) (t^{\beta/\alpha}\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\ &= t^{2n(\beta+\gamma-1)+2n} \int_{\mathbb{R}^{nd}} \left(\int_{[0,1]^{n <}} \overline{\mathcal{F}Y(1-t_n, \cdot) (t^{\beta/\alpha}(\xi_1 + \cdots + \xi_n)) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (t^{\beta/\alpha}\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\ &= t^{2n(\beta+\gamma-1)+2n} \int_{\mathbb{R}^{nd}} \left(\int_{[0,1]^{n <}} \overline{\mathcal{F}Y(1-t_n, \cdot) (\xi_1 + \cdots + \xi_n) \cdots \mathcal{F}Y(t_2 - t_1, \cdot) (\xi_1)} \right)^2 \varphi(t^{-\beta/\alpha}\vec{\xi}) t^{-nd\beta/\alpha} d\vec{\xi} \end{aligned}$$

$$\begin{aligned}
&= t^{2n(\beta+\gamma-1)+2n-n\beta\Delta/\alpha} \int_{\mathbb{R}^{nd}} \left(\int_{[0,1]^n} \overline{\mathcal{F}Y(1-t_n, \cdot)(\xi_1 + \dots + \xi_n) \cdots \mathcal{F}Y(t_2 - t_1, \cdot)(\xi_1)} \right)^2 \varphi(\vec{\xi}) d\vec{\xi} \\
&= t^{2n(\beta+\gamma-1)+2n-n\beta\Delta/\alpha} \int_{\mathbb{R}^{nd}} \mathcal{F}f_n(\cdot, 0, 1)(\vec{\xi})^2 \varphi(\vec{\xi}) d\vec{\xi} \\
&= t^{[2(\beta+\gamma-1)+2-\beta\Delta/\alpha]n} \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{H^{\otimes n}}^2
\end{aligned}$$

Nick's Question 2.2 **Lemma 2.3** We define

$$\mathcal{F}_a = \left\{ g \in H^a(\mathbb{R}^d) : \|g\|_{L^2(\mathbb{R}^d)} = 1, \mathcal{E}_a(g, g) < \infty \right\}$$

and then we also define

$$\mathcal{M}_a(f, \theta) = \sup_{g \in \mathcal{F}_a} \left\{ \langle g^2 * f, g^2 \rangle^{1/2} - \frac{\theta}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi \right\} \quad (2.1) \quad \text{var}$$

Now we define $\tilde{g}(x) = B^{-d/2}g(x/B)$. Its easy to show that $\mathcal{F}\tilde{g}(\xi) = B^{d/2}\mathcal{F}g(B\xi)$. By a change of variables, we can then show that

$$\int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi = B^{-a} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}g(\xi)|^2 d\xi$$

Next, by applying the scaling property of f , $f(cx) = c^{-\alpha}f(x)$, we can show by again using a change of variables that

$$\begin{aligned}
(\tilde{g}^2 * f)(x) &= B^{-d} \int_{\mathbb{R}^d} \tilde{g}^2(y/B) f(x-y) dy \\
&= \int_{\mathbb{R}^d} g^2(u) f(x-uB) du \\
&= B^{-\alpha} \int_{\mathbb{R}^d} g^2(u) f\left(\frac{x}{B} - u\right) du \\
&= B^{-\alpha} (g^2 * f)(x/B)
\end{aligned}$$

This can be used to show that

$$\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{-\alpha} \langle g^2 * f, g^2 \rangle$$

Now we can rewrite (2.1) as the following

$$\mathcal{M}_a(\Theta f, \theta) = \frac{1}{\theta} \sup_{g \in \mathcal{F}_a} \left\{ \frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} \langle \tilde{g}^2 * f, \tilde{g}^2 \rangle^{1/2} - \frac{B^a}{2} \int_{\mathbb{R}^d} |\xi|^a |\mathcal{F}\tilde{g}(\xi)|^2 d\xi \right\}$$

and as before, in order to preserve a scaling property, we want to choose B such that

$$\frac{\sqrt{\Theta}}{\theta} B^{\alpha/2} = B^a$$

and solving for B we get

$$B = \left(\frac{\sqrt{\Theta}}{\theta} \right)^{\frac{2}{2a-\alpha}}$$

and after plugging this B in we get that

$$\mathcal{M}_a(\Theta f, \theta) = \Theta^{\frac{a}{2a-\alpha}} \theta^{-\frac{\alpha}{2a-\alpha}}$$